

field ζ on a manifold M is

$$\begin{aligned}\zeta : M &\rightarrow TM \\ x &\mapsto \zeta_x\end{aligned}$$

$$+ \dots + X_n \partial_n$$

$$(\dots, X_n)$$

is a vector field on M .

Tensor $\otimes$ $\otimes$

What is a tensor

Definition

$$f(x, y)$$

$$\text{if } f = \sum_{i,j} a_{ij} x_i x_j$$

$$x = (x_1, \dots, x_n)$$

Tensor maps : $f \mapsto f \circ \phi$

$$(\phi)_x : F \rightarrow F$$

$$f \mapsto f \circ \phi$$

It satisfies

$$R_{\phi} = \phi^*$$

$$f \mapsto f \circ \phi$$



Differential

Up to now, we can only compute differential of a mapping defined on an Euclidean space.

What if the mapping is defined on a manifold M ?

$F : M \rightarrow N$ smooth mapping between two manifolds

ζ_x : tangent vector at $x \in M$

Define the mapping $DF(x)$ as

$$(DF(x)[\zeta])f := \zeta(f \circ F), \quad f \in \mathcal{F}(N)$$

it is a tangent vector to N at $F(x)$.

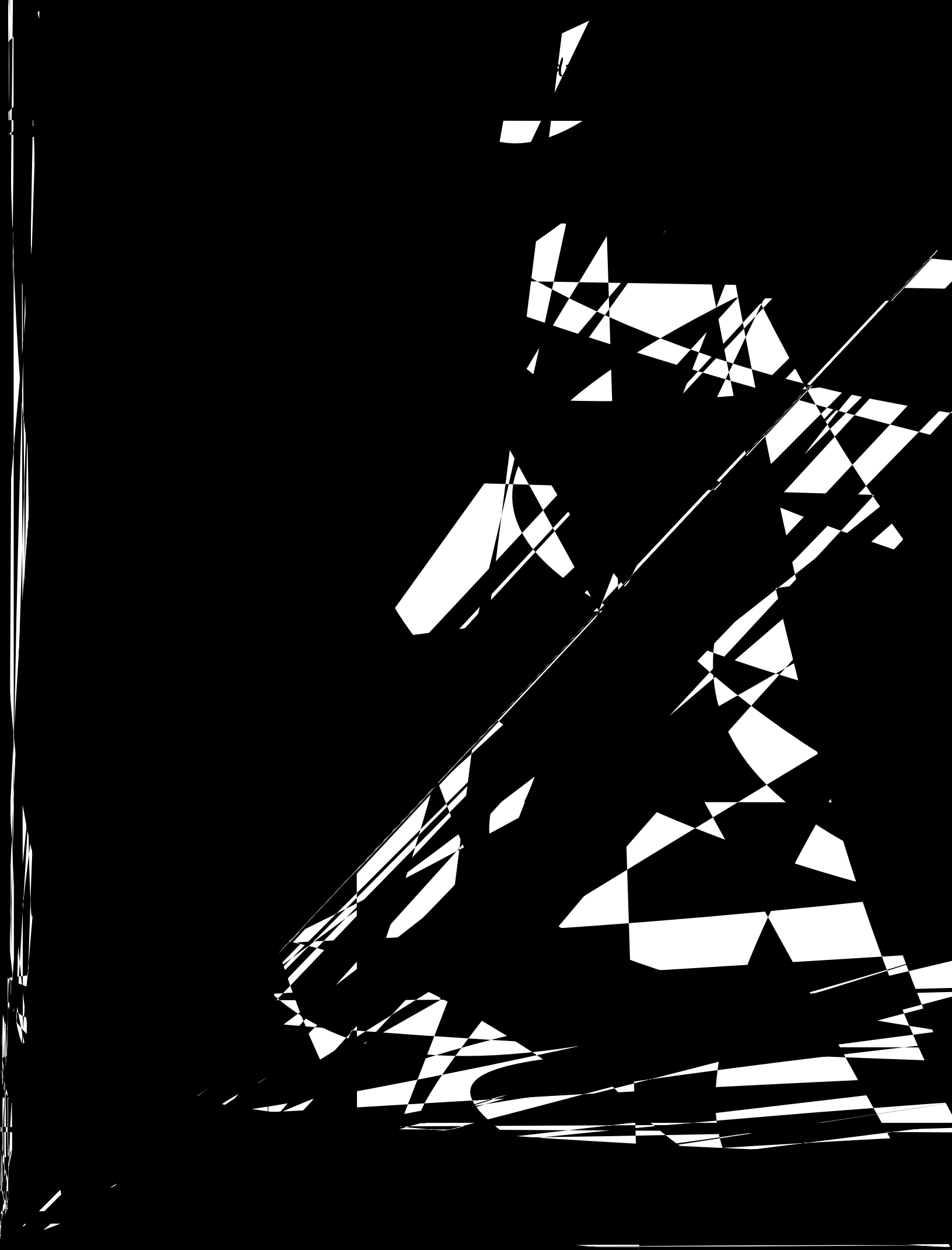
(If ζ_x is realized by some curve γ in M .

then $DF(x)[\zeta]$ is realized by the curve $F \circ \gamma$ in N .)

This mapping

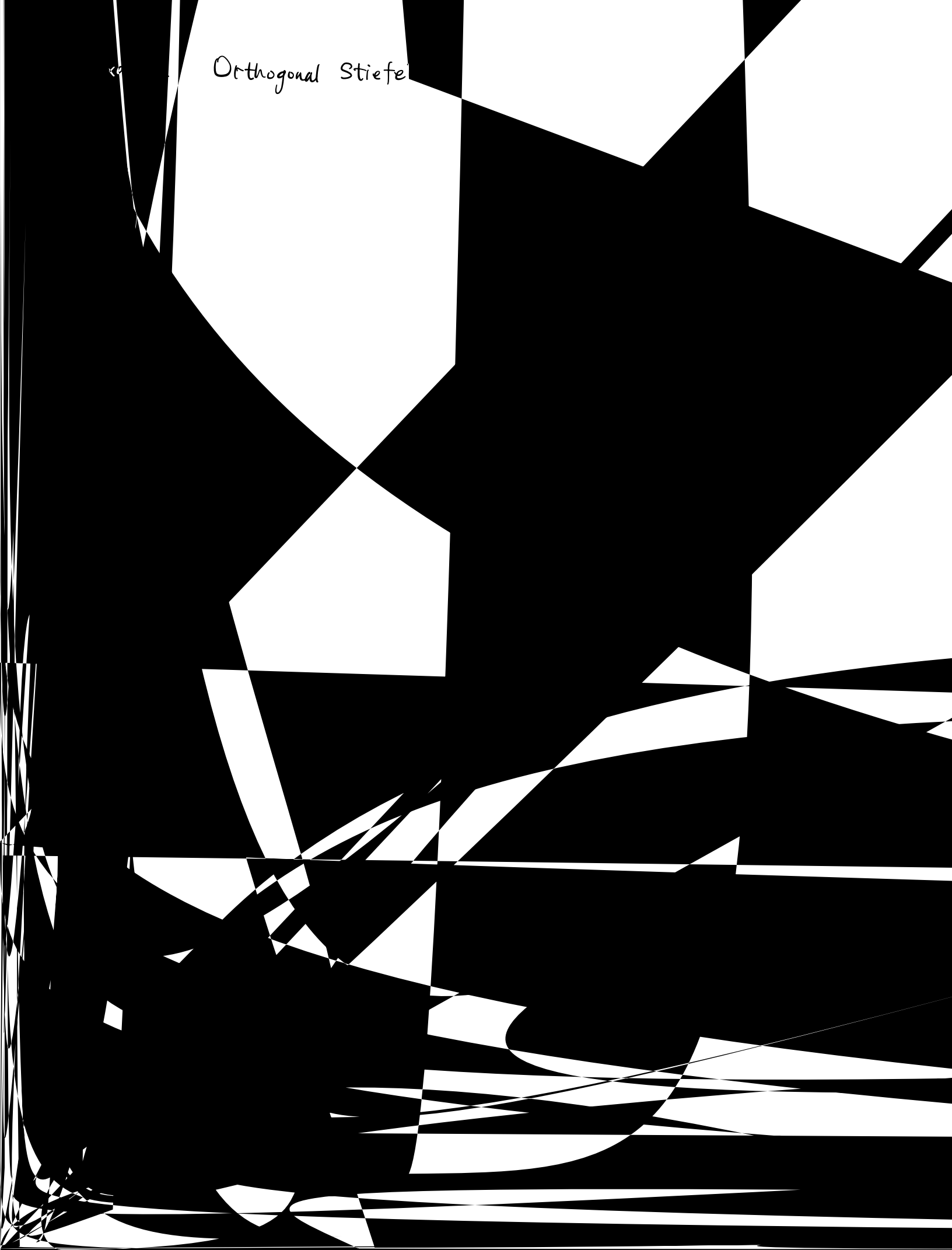
$$\begin{aligned} DF(x) : T_x M &\longrightarrow T_{F(x)} N \\ \zeta &\longmapsto DF(x)[\zeta] \end{aligned}$$

is a linear mapping, called the differential (differential map, derivative, or tangent map) of F at x .





Orthogonal Stiefe



Method

$$X = C^T X$$

that $\dim Z = 2$

$$S = F^{-1}(I_p)$$

$$\text{Ker}(I)$$

$$\{Z \in F$$

$$Z$$

Method 3

Native $X(t)$ has full co

$$X_L(t)$$

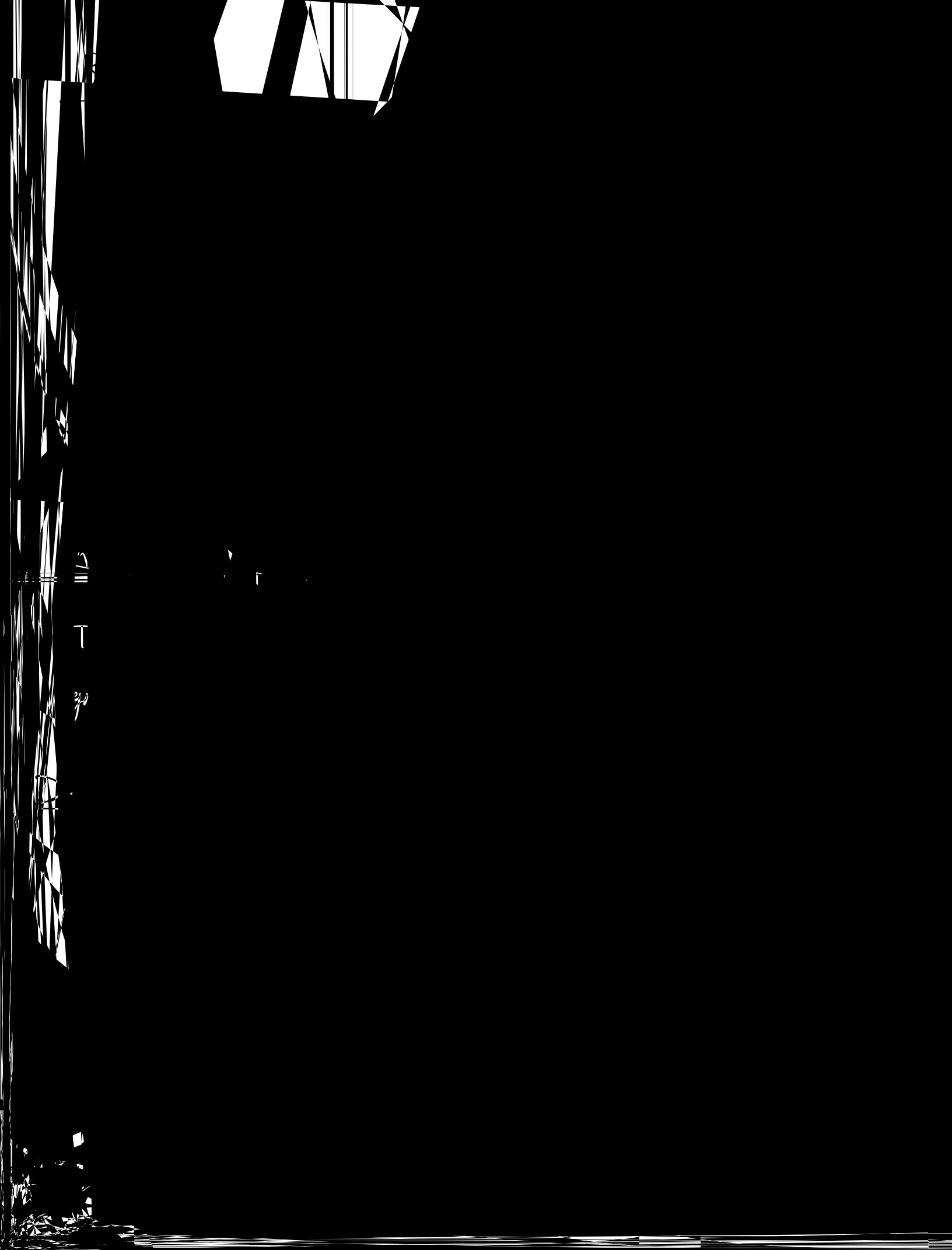
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$$\text{an}(X_L(t))$$

Example 3 (Orthogonal group)

As a special case O_n

$$T_0 O_n = \{Z \in \mathbb{R}^{n \times n} \mid Z^T = -Z\}$$





general theory of decomposition-based retractions.

M is an embedded manifold of a vector space E .

$\mathcal{N}$ is an abstract manifold s.t. $\dim(M) + \dim(\mathcal{N}) = \dim(E)$.

Assume $\phi: \mathcal{N} \times E \rightarrow E$ is a diffeomorphism

on $\mathcal{N} \times \mathcal{E}^*$ where $\mathcal{E}^* \subset E$ is an open subset of E .

$$\phi(F, G) = \phi(F, G)$$

with a neutral element $I \in \mathcal{N}$ s.t

$$\phi(F, I) = F, \quad \forall F \in \mathcal{N}$$

So Under the above assumptions, the mapping

$$R_X(\mathcal{Z}) := \pi_1(\phi(\mathcal{Z}, \mathcal{Z}))$$

where $\pi_1: M \times \mathcal{N} \rightarrow M$

$$(F, G) \mapsto F$$

the projection onto the first component,

defines a retraction on M .

$$F_0 \in \mathcal{E}^* \text{ open} \Rightarrow \forall \mathcal{Z} \in \mathcal{E}^* \quad \phi(\mathcal{Z}, \mathcal{Z}) \in \mathcal{E}^* \quad \text{if } \mathcal{Z} \in F_0$$

$\Rightarrow R_X(\zeta)$ defined $\forall \zeta$ in a neighborhood of the origin $T_X \mathcal{M}$.

R is smooth

$$R_X(0_X) = X \quad \text{direct}$$

Local rigidity : $\forall \zeta \in T_X \mathcal{M}$.

$$D_1 \phi(X, I)[\zeta] = D\phi(X, I)[(\zeta, 0)] = \zeta$$

Since $\pi_* \circ \phi^{-1}(\phi(F, I)) = F$, $\forall F \in \mathcal{M}$, it follows that

if we take the derivative at $X \in \mathcal{M}$ in the direction ζ :

$\forall \zeta \in T_X \mathcal{M}$.

$$D \text{id}_{\mathcal{M}}(X)[\zeta] = D(\pi_* \circ \phi^{-1})(\phi(X, I))[D_1 \phi(X, I)[\zeta]]$$

$$\text{Use } \phi(X, I) = X, \quad D_1 \phi(X, I)[\zeta] = \zeta$$

$$\Rightarrow \zeta = D(\pi_* \circ \phi^{-1})(X)[\zeta] = DR_X(0_X)[\zeta]$$

which proves that R_X is a contraction.

□

Method 1 - Gram-Schmidt decomposition

$$QR = A \quad Q \in \mathbb{R}^{n \times n}$$

$$R \in \mathbb{R}^{n \times n}$$

$$A = QR$$

$$R = \begin{bmatrix} R_{\text{supp}} & 0 \\ 0 & I_{\text{supp} + (n)} \end{bmatrix}$$

Define ϕ as the inverse of ψ



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Method 2: (Polar decomposition)

$$A = QP$$

$Q \in O_n$, $P \in S_{\text{sym}+}(n)$: all symmetric positive definite matrices of size n

The inverse of polar decomposition is

$$\begin{aligned}\phi : O_n \times S_{\text{sym}+}(n) &\rightarrow \mathbb{R}_{*}^{n \times n} \\ (Q, P) &\mapsto QP\end{aligned}$$

$$\text{We have } \phi^{-1}(A) = (A(A^T A)^{-\frac{1}{2}}, (A^T A)^{\frac{1}{2}})$$

This shows that ϕ is a diffeomorphism, thus

$$\begin{aligned}R_X(X\Omega) &:= X(I+\Omega) \left[(X(I+\Omega))^T (X(I+\Omega)) \right]^{-\frac{1}{2}} \\ &= X(I+\Omega) (I-\Omega^2)^{-\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\text{Note: } & (X(I+\Omega))^T (X(I+\Omega)) \\ &= (I+\Omega)^T X^T X (I+\Omega) \\ &= (I+\Omega)^T (I+\Omega) \quad (X \text{ orthogonal}) \\ &= (I-\Omega)(I+\Omega) \quad (\Omega \text{ skew-symmetric})\end{aligned}$$

Computing this function requires an eigenvalue decomposition of the $n \times n$ symmetric matrix $(I-\Omega^2)$.



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